Supplementary material

Mesh-to-Mesh Transfer Errors in Multistep Unsteady Flow Simulations

This supplement contains the full Euler remeshing comparisons and additional tests of the transferred history states supporting the article. The tables also give the convergence controls for the viscous airfoil case and a companion turbulence-model calculation. NN, BAR, GP, and CV denote nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and control-volume transfer; suffixes give nominal order of accuracy for smooth fields. GC denotes bounded global correction; LC denotes locally conservative flux correction. BDF2 denotes the second-order backward differentiation formula.

S1 Repeated remeshing in Euler simulations

Table S1 includes the higher-order alternatives to the main-text comparison, and Table S2 reports all three realizations, including globally corrected NN1. The remeshing interval is given in physical time steps. The deformation parameter s bounds streamwise displacement by the smallest incident corner-triangle altitude divided by s; $s = 2$ permits twice the displacement of $s = 4$. The wide-window position is the 50% crossing, and apparent width is the distance between the 10% and 90% crossings using plateaus in $W_1 = [2.40, 3.00]$. At 398 transfers, contact smearing contaminates the left plateau for NN1 and BAR2, so these widths are not shock-only measures. Differences from the fixed-mesh calculation are expressed in cells of spacing $h = 0.01$. For nodal density $\rho_{M,i}$ on mesh M, let $Q_M(\rho_M) = \sum_i w_i^M \rho_{M,i}$ with median-dual measure w_i^M, and write $I_\rho = Q_M(\rho_M)$. Then $\sum \Delta I_\rho / I_{\rho,0}$ is the signed sum of target-minus-source density-integral changes immediately across transfer, normalized by initial mass and excluding physical evolution. Integral cancellation leaves local distortion unconstrained, so the tables report both integral and profile errors.

Table S1: Frequency comparison at deformation parameter $s = 4$ for all transfer variants. Δn is the number of physical time steps between mesh changes; $\Delta x/h$ and $\Delta w^{\mathrm{wide}}/h$ are signed wide-window position and apparent-width changes relative to the fixed-mesh reference. Widths can include contact smearing and must not be read as shock thickness.

Transfer	Δn	$\Delta x/h$	$\Delta w^{\mathrm{wide}}/h$	$\sum \Delta I_\rho / I_{\rho,0}$
NN1	10	0.14	0.57	$1.1 \cdot 10^{-4}$
NN1	2	0.1	21.41	$5 \cdot 10^{-4}$
NN2	10	−0.02	−0.01	$-8.8 \cdot 10^{-6}$
NN2	2	−0.04	0	$-1.7 \cdot 10^{-5}$
NN3	10	0.01	−0.01	$2.2 \cdot 10^{-5}$
NN3	2	0.02	0	$2.7 \cdot 10^{-5}$
BAR2	10	0.01	0.25	$2 \cdot 10^{-5}$
BAR2	2	−0.03	3.24	$1.4 \cdot 10^{-4}$
BAR3	10	0	0.01	$-1.7 \cdot 10^{-5}$
BAR3	2	0	0.07	$-5.2 \cdot 10^{-6}$
GP2	10	0.01	0	$2.6 \cdot 10^{-6}$
GP2	2	0.01	0	$-2.8 \cdot 10^{-6}$
GP3	10	−0.01	0	$-2.1 \cdot 10^{-5}$
GP3	2	−0.01	0.02	$1.8 \cdot 10^{-5}$
CV2	10	0	0	$3 \cdot 10^{-14}$
CV2	2	0	0.01	$-8.2 \cdot 10^{-15}$

Table S2: High-frequency Euler comparison across three random deformation realizations. GC indicates whether the global correction is applied; $s = 2$ permits twice the displacement amplitude of $s = 4$. Position and width changes are in cells of the $h = 0.01$ reference mesh. Widths use W_1 and include contact smearing for strongly diffused cases. Transfer-integral diagnostics are available in the archived data.

Transfer	GC	s	Realization	$\Delta x/h$	$\Delta w^{\mathrm{wide}}/h$
BAR2	No	2	1	−0.06	22.91
BAR2	No	2	2	−0.03	22.98
BAR2	No	2	3	−0.07	22.9
BAR2	No	4	1	−0.03	3.24
BAR2	No	4	2	0	3.2
BAR2	No	4	3	−0.04	3.12
CV2	No	2	1	0.19	0.06
CV2	No	2	2	0.2	0.07
CV2	No	2	3	0.18	0.05
CV2	No	4	1	0	0.01
CV2	No	4	2	0.01	−0.01
CV2	No	4	3	0	0.01
GP2	No	2	1	0.17	0.06
GP2	No	2	2	0.19	0.06
GP2	No	2	3	0.19	0.03
GP2	No	4	1	0.01	0
GP2	No	4	2	0.02	0
GP2	No	4	3	0.01	0.01
NN1	Yes	2	1	−0.91	25.65
NN1	Yes	2	2	−0.65	23.8
NN1	Yes	2	3	0.01	26.33
NN1	Yes	4	1	−0.15	18.43
NN1	Yes	4	2	−0.01	21.28
NN1	Yes	4	3	−0.11	21.2
NN1	No	2	1	−0.38	26.36
NN1	No	2	2	−0.29	24.98
NN1	No	2	3	0.44	27.14
NN1	No	4	1	0.1	21.41
NN1	No	4	2	0.3	22.89
NN1	No	4	3	0.2	22.75

S2 Boundary residuals and the cancellation test

S2.1 First-step response ratios

The source mesh is A and the target mesh B. After the first BDF2 step, the state difference is $\varepsilon^{n+1} = \widehat{U}_B^{n+1} - U_B^{n+1}$, with both states computed on B. At the two input times, ε^{n-1} and ε^n denote the differences introduced by transferring the histories. Subscripts both, latest, and older identify which stored states were perturbed. All responses use the steady discrepancy $\varepsilon^{\mathrm{st}} = \mathcal{T}_{AB}(U_A^{\mathrm{st}}) - U_B^{\mathrm{st}}$, where the superscript st denotes the independently computed steady references and $\mathcal{T}_{AB}$ is the complete transfer including corrections. The lift difference from the target-mesh continuation is ΔC_L.

States are stacked vertex by vertex. The mass matrix is $M_B = \mathrm{diag}(w_i^B) \otimes I_{n_U}$, where w_i^B is the area of the target control volume around vertex i in this two-dimensional test, n_U counts the state components, and I_{n_U} is their identity matrix. The symbol $\otimes$ denotes the Kronecker product and a superscript T denotes transpose. The first-step state difference has volume-weighted norm $\|\varepsilon^{n+1}\|_{M_B} = [(\varepsilon^{n+1})^T M_B \varepsilon^{n+1}]^{1/2}$ and Euclidean norm $\|\varepsilon^{n+1}\|_2 = [(\varepsilon^{n+1})^T \varepsilon^{n+1}]^{1/2}$.

The superposition remainders are $\mathcal{E}_U = \varepsilon^{n+1}_{\mathrm{both}} - \varepsilon^{n+1}_{\mathrm{latest}} - \varepsilon^{n+1}_{\mathrm{older}}$ and $\mathcal{E}_{C_L} = \Delta C_{L,\mathrm{both}} - \Delta C_{L,\mathrm{latest}} - \Delta C_{L,\mathrm{older}}$. The relative errors are $e^U_{\sup} = \|\mathcal{E}_U\|_{M_B} / \|\varepsilon^{n+1}_{\mathrm{both}}\|_{M_B}$ and $e^{C_L}_{\sup} = |\mathcal{E}_{C_L}| / |\Delta C_{L,\mathrm{both}}|$. All reported denominators are nonzero.

For the same imposed error field, the BDF2 residual coefficients have exact ratios of $4/3$ and $-1/3$ relative to perturbing both histories. The first-step state and lift inherit these ratios to first order under the smoothness and solve assumptions in the article. The state-norm ratios are nonnegative, whereas lift ratios retain their signs.

Table S3: Euler BDF2 response ratios and relative superposition errors. Subscripts both, latest, and older identify the perturbed histories at the same output time. Linear response to the imposed error field predicts ratios of $4/3$ and $-1/3$; state columns show nonnegative norm ratios.

Transfer	$e^U_{\sup}$	$\frac{\|\varepsilon^{n+1}_{\mathrm{latest}}\|_{M_B}}{\|\varepsilon^{n+1}_{\mathrm{both}}\|_{M_B}}$	$\frac{\|\varepsilon^{n+1}_{\mathrm{older}}\|_{M_B}}{\|\varepsilon^{n+1}_{\mathrm{both}}\|_{M_B}}$	$e^{C_L}_{\sup}$	$\frac{\Delta C_{L,\mathrm{latest}}}{\Delta C_{L,\mathrm{both}}}$	$\frac{\Delta C_{L,\mathrm{older}}}{\Delta C_{L,\mathrm{both}}}$
NN1	$1.8 \cdot 10^{-2}$	1.3333	0.3335	$1.5 \cdot 10^{-2}$	1.3329	−0.3179
NN2	$3.5 \cdot 10^{-2}$	1.3331	0.3341	$4.2 \cdot 10^{-3}$	1.3318	−0.336
NN3	$3.4 \cdot 10^{-2}$	1.3331	0.334	$5.3 \cdot 10^{-3}$	1.3309	−0.3362
BAR2	$3.4 \cdot 10^{-2}$	1.3332	0.334	$3.1 \cdot 10^{-3}$	1.3328	−0.3358
BAR3	$3.3 \cdot 10^{-2}$	1.3332	0.334	$2.8 \cdot 10^{-3}$	1.3329	−0.3357
GP2	$3.4 \cdot 10^{-2}$	1.333	0.334	$3 \cdot 10^{-3}$	1.3323	−0.3353
GP3	$3.4 \cdot 10^{-2}$	1.3331	0.334	$3.1 \cdot 10^{-3}$	1.3327	−0.3358
CV2	$3.4 \cdot 10^{-2}$	1.333	0.334	$3.1 \cdot 10^{-3}$	1.3323	−0.3354

S2.2 Slip-wall comparison

When both residuals are evaluated at the same estimate of the new state U_B^{n+1}, the predicted residual perturbation is $-M_B \eta^n / \Delta t$, with $\eta^n = 2\varepsilon^n - \varepsilon^{n-1}/2$ formed from the errors in the latest and older states relative to the target-mesh reference. Here n labels physical time steps, ε^k is the restarted state minus the reference state on B at level k, and Δt is the physical step size. The solver prints the negative of the left-hand side of the BDF2 equation, so the predicted difference in its printed values is $+M_B \eta^n / \Delta t$. The comparisons use residual evaluations at a common field. They isolate the history contribution by keeping the estimate of the new state fixed, as specified by the common-trial-state simplification in the article. These diagnostics check the history forcing; Proposition 1 retains the different computed new states and their spatial residuals.

At the 370 slip-wall vertices, the wall-normal momentum mismatch is 3.4×10^{-12} for GP2, 4.1×10^{-12} for GP3 and CV2, and 4.5×10^{-12} for NN1. The tangential mismatch is $(1.4\text{–}1.7) \times 10^{-11}$, a fraction $(1.1\text{–}1.3) \times 10^{-7}$ of the tangential residual perturbation of 1.25×10^{-4}. All four transfers return the same wall state, that of the source solution; the difference in wall-normal

momentum from the target-mesh reference is 1.1×10^{-14}, so the normal residual perturbation is negligible and enforcement of the target slip-wall condition does not alter it. The predicted and measured perturbations both include enforcement of the target slip-wall condition.

S2.3 Residual probes in the cancellation test

In the cancellation test, the history discrepancies are $\varepsilon^n = \varepsilon^{\mathrm{st}}/4$ and $\varepsilon^{n-1} = \varepsilon^{\mathrm{st}}$, giving $\eta^n = 2\varepsilon^n - \varepsilon^{n-1}/2 = 0$. In this test the reference and restarted runs start their inner solves from different estimates of the flow at t^{n+1}. We record the constant-step BDF2 residual Φ_B before the first update in each run. Write $U_{B,\mathrm{trial}}^{n+1}$ for the reference initial guess and $\widehat{U}_{B,\mathrm{trial}}^{n+1}$ for the restarted initial guess. Their difference is $\varepsilon_{\mathrm{trial}}^{n+1} = \widehat{U}_{B,\mathrm{trial}}^{n+1} - U_{B,\mathrm{trial}}^{n+1}$. In the expression below, F_B is the spatial residual. With $\widehat{U}_B^k = U_B^k + \varepsilon^k$ at the two stored levels $k = n, n-1$, the residual difference is

$$\begin{aligned}\Delta\Phi_B &\equiv \Phi_B(\widehat{U}_{B,\mathrm{trial}}^{n+1}; \widehat{U}_B^n, \widehat{U}_B^{n-1}) - \Phi_B(U_{B,\mathrm{trial}}^{n+1}; U_B^n, U_B^{n-1}) \\ &= \frac{M_B}{\Delta t}\left(\tfrac{3}{2}\varepsilon_{\mathrm{trial}}^{n+1} - \eta^n\right) + F_B(\widehat{U}_{B,\mathrm{trial}}^{n+1}, t^{n+1}) - F_B(U_{B,\mathrm{trial}}^{n+1}, t^{n+1}).\end{aligned} \tag{S1}$$

With $\varepsilon_{\mathrm{trial}}^{n+1} = \varepsilon^{\mathrm{st}}/4$ and $\eta^n = 0$, the temporal part is three quarters of the magnitude of the residual perturbation with only the older state in error. The measured ratio near 0.75 also includes the change in F_B between the two initial guesses. Evaluating both residuals at the same estimate would give $\varepsilon_{\mathrm{trial}}^{n+1} = 0$, cancel the spatial terms and leave exactly $-M_B\eta^n/\Delta t$. Cancellation is temporary even for an exact first step: if the first-step state difference vanishes, the older level retains $\varepsilon^{\mathrm{st}}/4$, producing the subsequent history combination $-\varepsilon^{\mathrm{st}}/8$.

The computed lift differences in the cancellation test are of order 10^{-9}, but volume-weighted state differences remain 4.3–9.0% of the imposed discrepancy. Table S4 gives both state norms. Iterative error is not independently quantified, so this test does not verify cancellation of the computed state response.

Table S4: Cancellation of the initial residual perturbation. Both history states are perturbed but $2\varepsilon^n - \varepsilon^{n-1}/2 = 0$. First-step force differences and state differences use the same reference continuation; both state norms are shown to avoid interpreting a small load difference as complete state agreement.

Transfer	ΔC_L	ΔC_D	$\lVert\varepsilon^{n+1}\rVert_2/\lVert\varepsilon^{\mathrm{st}}\rVert_2$	$\lVert\varepsilon^{n+1}\rVert_{M_B}/\lVert\varepsilon^{\mathrm{st}}\rVert_{M_B}$
CV2	$7.59 \cdot 10^{-9}$	$-5.54 \cdot 10^{-9}$	$2.43 \cdot 10^{-4}$	$8.98 \cdot 10^{-2}$
GP2	$7.64 \cdot 10^{-9}$	$-5.54 \cdot 10^{-9}$	$2.43 \cdot 10^{-4}$	$8.98 \cdot 10^{-2}$
NN1	$2.11 \cdot 10^{-9}$	$-4.1 \cdot 10^{-9}$	$2.1 \cdot 10^{-4}$	$4.34 \cdot 10^{-2}$

S3 Perturbation-amplitude dependence

The state and lift remainders and both state norms are defined in Section S2. Scaling the same discrepancy $\varepsilon^{\mathrm{st}}$ by $\lambda = 1, 1/2, 1/4$ reduces the Euclidean and lift remainders for BAR2, GP2, and CV2 (Table S5). Here λ is the imposed amplitude factor. The absolute Euclidean remainder decreases by factors of about 3.2–3.5 per halving. The absolute volume-weighted remainder decreases more slowly, from approximately 2.4×10^{-5} to 1.5–1.7×10^{-5}, while its relative value rises from about 3.4% to 8.7–9.2%. These finite-amplitude tests do not establish the quadratic prediction uniformly in the reported norms. The remaining iterative error was not independently measured.

Table S5: Amplitude dependence of first-step superposition. $\mathcal{E}_U = \varepsilon^{n+1}_{\text{both}} - \varepsilon^{n+1}_{\text{latest}} - \varepsilon^{n+1}_{\text{older}}$ is the state remainder and $\mathcal{E}_{C_L}$ its lift counterpart. Relative remainders use the corresponding response with both states perturbed. The Euclidean and volume-weighted state metrics have different amplitude dependence.

Transfer	λ	$\|\mathcal{E}_U\|_2$	$\|\mathcal{E}_U\|_2/\|\varepsilon^{n+1}_{\text{both}}\|_2$	$\|\mathcal{E}_U\|_{M_B}$	$\|\mathcal{E}_U\|_{M_B}/\|\varepsilon^{n+1}_{\text{both}}\|_{M_B}$	$\lvert\mathcal{E}_{C_L}\rvert$
BAR2	1	$1.76 \cdot 10^{-3}$	$7.78 \cdot 10^{-3}$	$2.36 \cdot 10^{-5}$	$3.36 \cdot 10^{-2}$	$2.22 \cdot 10^{-5}$
BAR2	0.5	$5.45 \cdot 10^{-4}$	$4.82 \cdot 10^{-3}$	$1.97 \cdot 10^{-5}$	$5.6 \cdot 10^{-2}$	$4.74 \cdot 10^{-6}$
BAR2	0.25	$1.6 \cdot 10^{-4}$	$2.83 \cdot 10^{-3}$	$1.54 \cdot 10^{-5}$	$8.74 \cdot 10^{-2}$	$1.76 \cdot 10^{-6}$
GP2	1	$1.86 \cdot 10^{-3}$	$8.23 \cdot 10^{-3}$	$2.45 \cdot 10^{-5}$	$3.44 \cdot 10^{-2}$	$2.23 \cdot 10^{-5}$
GP2	0.5	$5.82 \cdot 10^{-4}$	$5.15 \cdot 10^{-3}$	$2.06 \cdot 10^{-5}$	$5.77 \cdot 10^{-2}$	$4.42 \cdot 10^{-6}$
GP2	0.25	$1.65 \cdot 10^{-4}$	$2.93 \cdot 10^{-3}$	$1.65 \cdot 10^{-5}$	$9.24 \cdot 10^{-2}$	$1.58 \cdot 10^{-6}$
CV2	1	$1.86 \cdot 10^{-3}$	$8.22 \cdot 10^{-3}$	$2.42 \cdot 10^{-5}$	$3.4 \cdot 10^{-2}$	$2.27 \cdot 10^{-5}$
CV2	0.5	$5.82 \cdot 10^{-4}$	$5.15 \cdot 10^{-3}$	$2.06 \cdot 10^{-5}$	$5.77 \cdot 10^{-2}$	$4.53 \cdot 10^{-6}$
CV2	0.25	$1.66 \cdot 10^{-4}$	$2.95 \cdot 10^{-3}$	$1.64 \cdot 10^{-5}$	$9.16 \cdot 10^{-2}$	$1.63 \cdot 10^{-6}$

S4 Inner convergence in the viscous airfoil calculations

The Spalart–Allmaras (SA) and shear-stress-transport (SST) calculations use BDF2 equations solved by pseudo-time iteration at each physical time step. The stopping target is $\log_{10} \mathrm{rms}(r_\rho) \leq -12$ where r_ρ is the nodal density-equation residual in solver normalization and rms denotes its root-mean-square over the solver vertices. Reaching the iteration limit alone does not establish convergence. Table S6 gives the SA records behind the main-article table and the companion SST records; final residuals are medians over physical time steps, not residuals of averaged solutions. The nominal rows cover the full remeshing interval and other rows the shorter sensitivity interval; Δt_0 denotes the nominal physical time step.

Table S6: Iterative convergence by method for the first deformation realization. Residuals are median final $\log_{10} \mathrm{rms}(r_\rho)$; $N_{\max}$ is the iteration limit.

Closure	Transfer	$N_{\max}$	$\Delta t/\Delta t_0$	Physical steps	Steps at limit	Residual
SA	GP2	800	1	398	152	-12.00
SA	NN2	800	1	398	155	-12.00
SA	CV2	800	1	398	147	-12.00
SA	GP2	1600	1	38	0	-12.00
SA	NN2	1600	1	38	0	-12.00
SA	CV2	1600	1	38	0	-12.00
SA	GP2	800	1/2	78	0	-12.00
SA	NN2	800	1/2	78	0	-12.00
SA	CV2	800	1/2	78	0	-12.00
SA	GP2	800	1/4	158	0	-12.00
SA	NN2	800	1/4	158	0	-12.00
SA	CV2	800	1/4	158	0	-12.00
SST	GP2	800	1	398	333	-11.10
SST	NN2	800	1	398	324	-11.24
SST	CV2	800	1	398	336	-11.11
SST	GP2	1600	1	38	28	-11.33
SST	NN2	1600	1	38	28	-11.33
SST	CV2	1600	1	38	30	-11.27
SST	GP2	6400	1	38	28	-11.27
SST	NN2	6400	1	38	30	-11.17
SST	CV2	6400	1	38	29	-11.18
SST	GP2	800	1/2	78	56	-11.27
SST	NN2	800	1/2	78	58	-11.32
SST	CV2	800	1/2	78	60	-11.23

S4.1 Sensitivity to the inner-iteration limit

For a scalar history $q(t)$, its RMS at N_s common physical times t_i is $\mathcal{R}(q) = (N_s^{-1} \sum_{i=1}^{N_s} q(t_i)^2)^{1/2}$. Let q_{nom} and q_{test} be the histories at the nominal and modified numerical settings. Their relative difference is

$$S_{\mathrm{test}}(q) = \frac{\mathcal{R}(q_{\mathrm{test}} - q_{\mathrm{nom}})}{\mathcal{R}(q_{\mathrm{nom}})}. \tag{S2}$$

The denominator is nonzero for all histories reported here. The individual responses are $\Delta C_{Q,m} = C_{Q,m} - C_{Q,\mathrm{ref}}$, where Q labels a monitored coefficient, m identifies the transfer method, and the reference is the fixed-mesh calculation at the same numerical setting. The difference between NN2 and GP2 is $P_Q = \Delta C_{Q,\mathrm{NN2}} - \Delta C_{Q,\mathrm{GP2}}$. In this table, $Q = L, D$ selects lift or total drag. The table reports the largest sensitivity among the individual methods and the sensitivity of their smaller difference.

Table S7: Effect of increasing the inner-iteration limit, relative to the nominal setting. Values are $100 S_{\mathrm{test}}(q)$ from Eq. (S2); individual columns give the largest NN2, GP2, or CV2 sensitivity, and difference columns refer to the NN2–GP2 difference. All entries are percentages.

		Individual responses		NN2–GP2 difference	
Closure	Iteration limit	ΔC_L	ΔC_D	P_L	P_D
SA	1600	0.3	0.2	10.9	8.4
SST	1600	4.5	1.4	13.7	6.0
SST	6400	8.5	2.6	67.7	25.5

For SA at successively smaller time steps, the RMS lift differences between methods are approximately 1.21×10^{-5}, 1.60×10^{-5}, and 2.08×10^{-5}. Drag values are 2.19×10^{-6}, 3.77×10^{-6}, and 4.55×10^{-6}. Individual RMS amplitudes are closer, but the differences between the time histories remain unresolved. SA quarter-step friction-drag history differences are approximately 76–92% of the nominal RMS response, so convergence of total drag does not establish convergence of each component.

Companion SST calculation The SST runs use the same airfoil, mesh sequences and flow conditions as the SA runs. Both stored histories include ρk and $\rho\omega$, where k is turbulent kinetic energy and ω is specific dissipation rate. Across the two mesh sequences the largest relative integral changes for NN2 are 52 and 37 times those for GP2 for ρk and $\rho\omega$, respectively. The integrals were recorded before the final lower bounds were imposed on k and ω.

The iteration record limits the interpretation of the SST loads. With 6400 inner iterations, GP2, NN2 and CV2 still reach the limit on 28, 30 and 29 of the 38 nominal-size steps, respectively (Table S6). Raising the limit from 800 to 6400 changes the NN2–GP2 lift difference by 67.7% of its nominal RMS (Table S7). Halving the physical step changes the NN2–GP2 drag difference by 110.3%, and 56–60 of the resulting 78 steps still reach the 800-iteration limit. The SST histories thus document sensitivity to the inner solve as well as to the physical step.

S5 Flux correction in repeated Euler transfers

Table S8 reports all 36 matched comparisons. Standard and LC cases use the same mesh trajectories. Position differences use the wide-window 50% crossing; widths in this table instead use the shock-only interval $[2.55, 2.85]$. Both are measured relative to the corresponding fixed-mesh reference in cells of spacing 0.01. The shock-only diagnostic uses the stored eight-line mean profile at 800 streamwise probes. At this time, the exact contact and shock are at approximately 2.370 and 2.699. For heavily smeared NN1/BAR2 profiles, contact diffusion can raise the wide-window

left plateau from approximately 0.266 to 0.287 and shift the 10% crossing into the contact tail. A large wide-window width therefore does not imply equally large shock broadening.

Table S8: Matched standard and LC transfers. Δn is the remeshing interval, s the deformation parameter, and the Realization column identifies the deformation realization. Subscripts std and LC identify standard transfer and local flux correction; the superscript shock distinguishes these widths from the wide-window widths in Section S1. Positions and shock-only widths are in reference cells.

Transfer	Δn	s	Realization	$\Delta x_{\mathrm{std}}/h$	$\Delta x_{\mathrm{LC}}/h$	$\Delta w_{\mathrm{std}}^{\mathrm{shock}}/h$	$\Delta w_{\mathrm{LC}}^{\mathrm{shock}}/h$
BAR2	10	4	1	0.01	0.01	0.14	0.14
BAR3	10	4	1	0	0	0	0
CV2	10	4	1	0	−0.01	−0.02	−0.02
GP2	10	4	1	0.01	−0.01	−0.01	−0.01
GP3	10	4	1	−0.01	−0.01	−0.03	−0.03
NN1	10	4	1	0.14	0.09	0.36	0.36
NN2	10	4	1	−0.02	−0.01	−0.03	−0.03
NN3	10	4	1	0.01	0	−0.03	−0.02
BAR2	2	4	1	−0.03	−0.02	1.14	1.13
BAR2	2	4	2	0	−0.01	1.02	1.02
BAR2	2	4	3	−0.04	−0.03	1.05	1.05
BAR3	2	4	1	0	0.02	0.08	0.07
CV2	2	4	1	0	−0.01	−0.01	−0.01
CV2	2	4	2	0.01	0	0.01	0.01
CV2	2	4	3	0	−0.01	−0.01	−0.01
GP2	2	4	1	0.01	−0.01	−0.01	−0.01
GP2	2	4	2	0.02	0	0	0.01
GP2	2	4	3	0.01	0	−0.01	0
GP3	2	4	1	−0.01	0	0.02	0.01
NN1	2	4	1	0.1	0.06	1.49	1.53
NN1	2	4	2	0.3	0.2	1.6	1.68
NN1	2	4	3	0.2	0.11	1.19	1.17
NN2	2	4	1	−0.04	−0.03	−0.02	−0.03
NN3	2	4	1	0.02	0	−0.01	−0.01
BAR2	2	2	1	−0.06	−0.06	2.28	2.28
BAR2	2	2	2	−0.03	−0.05	2.17	2.16
BAR2	2	2	3	−0.07	−0.07	2.17	2.17
CV2	2	2	1	0.19	0.01	0.01	0.01
CV2	2	2	2	0.2	0.01	0.03	0.05
CV2	2	2	3	0.18	0	−0.01	0.03
GP2	2	2	1	0.17	0.01	0.02	0.01
GP2	2	2	2	0.19	0.02	0.03	0.04
GP2	2	2	3	0.19	0.01	−0.02	0.03
NN1	2	2	1	−0.38	−0.53	8.68	8.68
NN1	2	2	2	−0.29	−0.34	8.37	8.34
NN1	2	2	3	0.44	0.29	8	8.03

The direct final drifts are normalized by the exact mass 2.375 and energy 5.75. Across all matched LC cases their absolute values are below 5×10^{-13} and 7×10^{-13}, respectively, in the double-precision restart after the final transfer. This is distinct from summing instantaneous transfer defects. Table S9 separates the final CFD-step and transfer changes for representative cases. A BDF2 calculation also retains a difference between its two history levels; the complete archived budget includes that quantity. The table entries describe individual stage changes, not an additive sum of the entire multistep history.

Table S9: Relative mass changes in the final CFD step and transfer, for realization 1 with remeshing every two steps. Standard is the usual transfer treatment; LC is the locally conservative flux correction. All values are normalized by initial mass.

Transfer	s	Treatment	CFD step	Transfer
NN1	4	Standard	$-3.5 \cdot 10^{-7}$	$-1.4 \cdot 10^{-5}$
NN1	4	LC	$1.7 \cdot 10^{-15}$	$1.9 \cdot 10^{-16}$
GP2	2	Standard	$1.1 \cdot 10^{-8}$	$4.6 \cdot 10^{-7}$
GP2	2	LC	$1.7 \cdot 10^{-15}$	$-1.9 \cdot 10^{-16}$
CV2	2	Standard	$2.7 \cdot 10^{-8}$	$0 \cdot 10^{0}$
CV2	2	LC	$0 \cdot 10^{0}$	$0 \cdot 10^{0}$

The density extrema permitted by LC are measured relative to the source range at each transfer. NN1 and BAR2 lose strict monotonicity, while excursions remain within the 2×10^{-3} bound relaxation. BAR2 reaches approximately 5.6×10^{-5}; the NN2, NN3, and BAR3 excursions are capped at 2×10^{-3}. The final flux-correction diagnostics reach the 12-pass limit in 23 of 36 cases. Remaining flux can therefore separate the returned state from the unlimited target even when conservation and the relaxed bounds are satisfied. No admissibility repair is activated. Wall-normal momentum at slip-wall vertices can reach 1.3×10^{-4} immediately after LC transfer and is removed by the next flow step. These results concern the specified Riemann configurations and do not establish an LC result for the RANS airfoil cases.

The standard and LC Riemann campaigns used transfer revisions `b3054414` and `0ac1b13`, respectively. The latter also changed the CV recovery solve and batched GP recovery. Although the prescribed mesh trajectories match, the comparison includes these implementation changes. The archive reports integral changes after corrections but does not provide a common set of achieved GP and CV linear residuals for all runs.

S6 Separating transfer error from mesh dependence

At either stored level $k = n-1, n$, let $U^k_{M,\mathrm{ex}}$ be the exact state sampled at the vertices of mesh M. The computed reference has discretization error $e^k_{\mathrm{disc},M} = U^k_M - U^k_{M,\mathrm{ex}}$. Adding and subtracting the transfer of the exact samples, $\mathcal{T}_{AB}(U^k_{A,\mathrm{ex}})$, gives the error in the transferred history

$$\varepsilon^k = \underbrace{\mathcal{T}_{AB}(U^k_{A,\mathrm{ex}}) - U^k_{B,\mathrm{ex}}}_{\text{transfer of the exact sampled field}} + \underbrace{\mathcal{T}_{AB}(U^k_A) - \mathcal{T}_{AB}(U^k_{A,\mathrm{ex}}) - e^k_{\mathrm{disc},B}}_{\text{source and target discretization contributions}} . \tag{S3}$$

For a linear transfer map, the second line reduces to $\mathcal{T}_{AB} e^k_{\mathrm{disc},A} - e^k_{\mathrm{disc},B}$. Limiting and admissibility corrections can make the complete map nonlinear.

For a scalar output functional G_M evaluated from the state on mesh M, a source-to-target output difference satisfies

$$\begin{aligned} G_B(\widehat{U}^k_B) - G_A(U^k_A) = {} & G_B(U^k_B) - G_A(U^k_A) \\ & + G_B(\widehat{U}^k_B) - G_B(U^k_B). \end{aligned} \tag{S4}$$

Here $\widehat{U}^k_B = \mathcal{T}_{AB}(U^k_A)$. The first term compares the two discrete mesh references. The second measures the output difference on B immediately after transfer.

S6.1 How the history weights change for a later load

Consider a load G_B at step $N \geq n+1$ on the same stationary target mesh, at constant Δt and with no further transfers. At each step m, define the spatial Jacobian at the reference state by $J_m = \partial_U F_B(U_B^m, t^m)$ and the scaled step Jacobian by $A_m = \frac{3}{2}M_B + \Delta t J_m$. The state difference is $\varepsilon^m = \widehat{U}_B^m - U_B^m$. Let $\mathfrak{r}_B^m$ and $\widehat{\mathfrak{r}}_B^m$ be the BDF2 residuals evaluated at each run's own state and histories when its inner solve stops. The spatial remainder is

$$\mathcal{N}_m(\varepsilon^m) = F_B(\widehat{U}_B^m, t^m) - F_B(U_B^m, t^m) - J_m\varepsilon^m.$$

The full difference equation at each advanced step is therefore

$$\begin{aligned} A_m\varepsilon^m = M_B(2\varepsilon^{m-1} - \tfrac{1}{2}\varepsilon^{m-2}) \\ + \Delta t(\widehat{\mathfrak{r}}_B^m - \mathfrak{r}_B^m) - \Delta t\,\mathcal{N}_m(\varepsilon^m). \end{aligned} \tag{S5}$$

This is the one-step BDF2 balance from the article applied at each time, with different new states and spatial residuals retained. The state and adjoint systems must use the same boundary constraints.

For the final output, define $\Delta G^N = G_B(\widehat{U}_B^N) - G_B(U_B^N)$, its reference gradient $g_N = \nabla_U G_B(U_B^N)$, and the output remainder $\mathcal{N}_G^N = \Delta G^N - g_N^T\varepsilon^N$. Set $g_m = 0$ for earlier steps. Assuming each A_m is invertible, start with $\boldsymbol{\lambda}_{N+1} = \boldsymbol{\lambda}_{N+2} = 0$ and solve backwards for $m = N, N-1, \ldots, n+1$:

$$A_m^T\boldsymbol{\lambda}_m = g_m + 2M_B\boldsymbol{\lambda}_{m+1} - \tfrac{1}{2}M_B\boldsymbol{\lambda}_{m+2}. \tag{S6}$$

The coefficients 2 and $-1/2$ come from the next two BDF2 steps. Multiply Eq. (S5) by $\boldsymbol{\lambda}_m^T$ and sum over $m = n+1, \ldots, N$. Equation (S6) cancels the intermediate state differences. Using $M_B^T = M_B$, the exact final-output balance is

$$\begin{aligned} \Delta G^N = (2\boldsymbol{\lambda}_{n+1} - \tfrac{1}{2}\boldsymbol{\lambda}_{n+2})^T M_B\varepsilon^n - \tfrac{1}{2}\boldsymbol{\lambda}_{n+1}^T M_B\varepsilon^{n-1} \\ + \Delta t \sum_{m=n+1}^{N} \boldsymbol{\lambda}_m^T(\widehat{\mathfrak{r}}_B^m - \mathfrak{r}_B^m) \\ - \Delta t \sum_{m=n+1}^{N} \boldsymbol{\lambda}_m^T\mathcal{N}_m(\varepsilon^m) + \mathcal{N}_G^N. \end{aligned} \tag{S7}$$

The first line contains the contributions of the two transferred histories. The remaining lines carry the inner-solve residual differences and the nonlinear remainders through to the final load. The adjoint weights include the spatial Jacobian at each reference new state.

For exact inner solves and small state differences on a common smooth solution branch, discarding the nonlinear remainders gives the leading history-weight formula

$$\Delta G^N \simeq (2\boldsymbol{\lambda}_{n+1} - \tfrac{1}{2}\boldsymbol{\lambda}_{n+2})^T M_B\varepsilon^n - \tfrac{1}{2}\boldsymbol{\lambda}_{n+1}^T M_B\varepsilon^{n-1}. \tag{S8}$$

The remainders are quadratic in the state differences under the gradient and Jacobian smoothness assumptions in the article. At a fixed finite horizon their adjoint-weighted sums determine the accuracy of this approximation. Early termination of the inner solves adds the residual sum in Eq. (S7). For $N = n+1$, only $\boldsymbol{\lambda}_{n+1}$ remains; Eqs. (S7) and (S8) recover the article's full one-step functional balance and its leading term, respectively. These expressions interpret the forward calculations; the reported tables do not evaluate the adjoint weights numerically.

S7 Steady mesh sensitivity and the medium-mesh transient

The steady meshes in Table S10 retain the first off-wall spacing while refining other regions. For SST, the coarse steady reference meets the density-residual target and the medium reference ends at $\log_{10}\mathrm{rms}(r_\rho) \simeq -11.1$. Lift C_L, total drag C_D, and pressure drag $C_{D,p}$ decrease monotonically; viscous drag $C_{D,v}$ does not. Let n_M be the vertex count of mesh M, and n_{coarse} that of the coarse mesh. Define $h_{\mathrm{eff},M} = n_M^{-1/2}$, with the coarse-mesh value defined in the same way. Their ratio $h_{\mathrm{eff},M}/h_{\mathrm{eff,coarse}} = \sqrt{n_{\mathrm{coarse}}/n_M}$ describes the density of this two-dimensional mesh family. It does not establish a common asymptotic error expansion or a refinement ratio in every direction. The table therefore reports absolute-load sensitivity without extrapolated loads or a discretization-uncertainty estimate. It does not quantify error in the load difference between transfer methods on a common mesh.

Table S10: Steady aerodynamic loads without remeshing on the coarse, medium, and fine viscous meshes. The vertex-count estimate $h_{\mathrm{eff},M}/h_{\mathrm{eff,coarse}} = \sqrt{n_{\mathrm{coarse}}/n_M}$ uses the coarse mesh as reference; it is not a refinement ratio common to all directions. The same first off-wall spacing is used on all levels. Entries retain the archived precision; rounded drag components can differ in the last digit from the displayed total.

Closure	Mesh	Vertices	$\frac{h_{\mathrm{eff},M}}{h_{\mathrm{eff,coarse}}}$	C_L	C_D	$C_{D,p}$	$C_{D,v}$
SA	Coarse	23,271	1	0.67626	0.010229	0.003559	0.006671
SA	Medium	32,102	0.851	0.67084	0.010048	0.003401	0.006647
SA	Fine	48,228	0.695	0.66854	0.009996	0.00333	0.006665
SST	Coarse	23,271	1	0.67561	0.009858	0.003469	0.006389
SST	Medium	32,102	0.851	0.6686	0.009705	0.003321	0.006385
SST	Fine	48,228	0.695	0.66639	0.009644	0.003247	0.006397

S7.1 Medium-mesh remeshing response

The medium-mesh SA test changes both resolution and the imposed nodal disturbance. Its short remeshing interval gives RMS lift responses near 1.07×10^{-4} and drag responses near 5.39×10^{-5} for individual methods. Differences between methods are 8.77×10^{-6} and 9.94×10^{-7}. The methods remain close, but the changed random displacement field prevents attributing the smaller response to resolution alone. Five to seven of the 38 physical steps reach the nominal inner-iteration limit.